

Derivative-Launch Transport Certificates for Finite Quantum Capture Windows

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Abstract—Direct propagation of derivative launches avoids inferring derivative accuracy from state accuracy. For the named MQM vacuum launch, three source vectors are p_0 , p_1 , and $p_0 + p_2$. We compare each finite Gauss reference with the continuum through a centered bath embedding, charge inherited directed coefficient and binary64 arithmetic allowances, and enclose the retained population slope and curvature. With inherited thermal-response and Gaussian bounds, rational arithmetic certifies retained error below $115/2^{23} < 2^{-16}$ on 16 ± 2^{-8} at unchanged $\beta = 5/16$. This is an eightfold wider halfwidth than the preceding launch-moment certificate. The result concerns a finite interval under the stated fixed-launch contract; permanent retention, arbitrary-input diamond accuracy, external review and hardware validation remain open.

Index Terms—Quantum control, continuum reservoirs, derivative propagation, rigorous error bounds.

I. FROZEN CONTRACT

Keep $\Omega = 64$, $B = 16$, $\kappa = 8$, $\beta = 5/16$, $T = 16$, $h = g = 1$, and $\lambda^2 = 4\sqrt{2}$. In the centered source frame the first live edges are $p_0 \leftrightarrow p_1 \leftrightarrow p_2$ with unit weights, followed by $p_2 \leftrightarrow p_3$ with weight λ . The band couples only to p_3 . The launch is $p_0 \otimes \text{vac}$ and P selects the four live coordinates.

The prior packet [1] established halfwidth 2^{-11} . Before fresh evaluation we declared candidates 2^{-k} , $k = 6, \dots, 24$, and horizon $U = 16 + 2^{-6} = 1025/64$. This is post-exposure engineering, not an independent prediction. Target, temperature and launch are unchanged. Mathematical modes are not physical qubits.

II. DIRECT CONTINUUM COMPARISON

For normalized launches the centered Gauss embedding of [2] gives a uniform interaction-picture port error $4GR$, where $G^2 = 128/\pi < 49$ and

$$R = \frac{x^{192}}{192!(1 - x/193)}, \quad x = BU/4 = 4U. \quad (1)$$

The conservative two-excitation Duhamel allowance also applies in the one-excitation sector:

$$D_1 \leq 8\sqrt{2}GUR < 84UR. \quad (2)$$

The embedding maps bath coordinates only and acts identically on the native coordinates. Separate free-bath unitaries leave their projection unchanged. Thus this bounds the native propagation discrepancy for each launch p_0 , p_1 , and $(p_0 + p_2)/\sqrt{2}$, without differentiating an approximate state

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comparison. For the unnormalized third vector multiply the allowance by $\sqrt{2} < 3/2$.

The exact source identities are $Hp_0 = p_1$ and $H^2p_0 = p_0 + p_2$. Let $u_j(t) = e^{-iHt}s_j$ for these three vectors. Then

$$q' = 2\Re\langle Pu_0, -iPu_1 \rangle, \quad (3)$$

$$q'' = 2\|Pu_1\|^2 - 2\Re\langle Pu_0, Pu_2 \rangle. \quad (4)$$

These follow from commutation of H with its propagator, not from numerical differentiation.

III. DIRECTED LAUNCH ENCLOSURES

At $M = 192$ we execute 128 Taylor steps of duration $1/8$ and degree 48, with real and imaginary CSR products separated. Returned binary64 entries are interpreted as exact rational data. Under the inherited round-to-nearest operation model [3], at most 196 row entries and $\|H\| < 64$ give the conservative normalized propagation allowance $256s$, with

$$s = 32 \cdot 2^{-43} 3^8 + \frac{8^{49}}{49!(1 - 8/50)} + 2^{-800}. \quad (5)$$

Use the inherited coefficient mismatch d_H from the directed parameter file. Its conservative two-excitation bound is valid here. The normalized vector debit is

$$e = 16d_H + 256s + 84UR < 2.138 \times 10^{-4}. \quad (6)$$

No normalized irrational launch preparation is required: the third seed has two exact unit entries, so its debit is bounded by $3e/2$.

Let a_j be rational upper roots of the exact returned native squared norms, and let $\tilde{q}'$ be the exact bilinear evaluation. The certified slope satisfies

$$|q'(16)| \leq |\tilde{q}'| + 2e(a_0 + a_1) + 2e^2 < 2^{-16}. \quad (7)$$

Here a_0 denotes the computed launch-zero norm only. In subsequent bounds use the separate inherited ceiling $A = 7/2048$, with $q(16) < E_0 < A^2$, $E_0 = 3/2^{18} + 2^{-128}$. The numerator 7 is unrelated to the proposed capacity coefficient $7/768$.

Since $\|Hp_1\| = \sqrt{2} < 3/2$ and $\|H(p_0 + p_2)\| = \sqrt{4 + 4\sqrt{2}} < 13/4$, for $|t - 16| \leq w$ we have

$$b_1(w) = a_1 + e + 3w/2, \quad (8)$$

$$b_2(w) = a_2 + 3e/2 + 13w/4, \quad (9)$$

$$C(w) = 2b_1(w)^2 + 2(A + w)b_2(w). \quad (10)$$

The live amplitude is unit-Lipschitz [1]. Consequently

$$|q(t) - q(16)| \leq 2^{-16}w + C(w)w^2/2. \quad (11)$$

At $w = 2^{-8}$, the curvature ceiling is approximately 5.431×10^{-4} ; this decimal is diagnostic, while the replay uses rational arithmetic.

IV. THERMAL WINDOW AND AUDIT

The inherited response point debit is $3/2^{26}$ and its finite variation is $2^{-20}w + w^2/8$. Uniform centered response comparison remains below 2^{-30} on $[0, U]$. The covariance comparison on this expanded horizon is below 2^{-56} ; its former 2^{-57} ceiling fails and is explicitly amended. Let $V = 64/31457265$,

$$x_w = \frac{12433696189517}{58512905602490430} + 6V(16w + w^2/2). \quad (12)$$

The Gaussian remainder is $x_w^2/[2(1 - x_w)] < 2^{-25}$. The total bound is the sum of E_0 , cold variation, response point and variation, both comparison debits, and this remainder.

Theorem 1. *Under the inherited continuum, Gaussian and arithmetic assumptions, retained thermal error is strictly below $115/2^{23} < 2^{-16}$ throughout $[16 - 2^{-8}, 16 + 2^{-8}]$.*

Proof. Exact rational replay gives a total diagnostic 1.348730×10^{-5} at $w = 2^{-8}$ and verifies the stated rational ceiling by integer cross multiplication. All variation terms majorize the closed interval. This is the largest passing member of the declared candidate list; 2^{-7} fails the bound, not the physical task. $\square$

Nine dense-eigensystem versus exponential-action controls at $M = 4, 8, 192$ verify both launch identities. Fresh Taylor output agrees with exponential action within 3.406×10^{-14} . Nineteen acceptance decisions are separately audited with integers. These same-agent routes are not external review. The failed initial display ceiling $113/2^{23}$ is preserved; changing the display ceiling did not change acceptance. Halfwidth increases eightfold, relative halfwidth is 2^{-12} and full width is 2^{-7} in model time. The equal numeral 2^{-16} in the slope ceiling and channel target does not identify their units or meanings.

The response variation now dominates enlargement. No device-time calibration, physical sink, perpetual retention, replenished reservoir or arbitrary-input channel guarantee follows.

REFERENCES

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